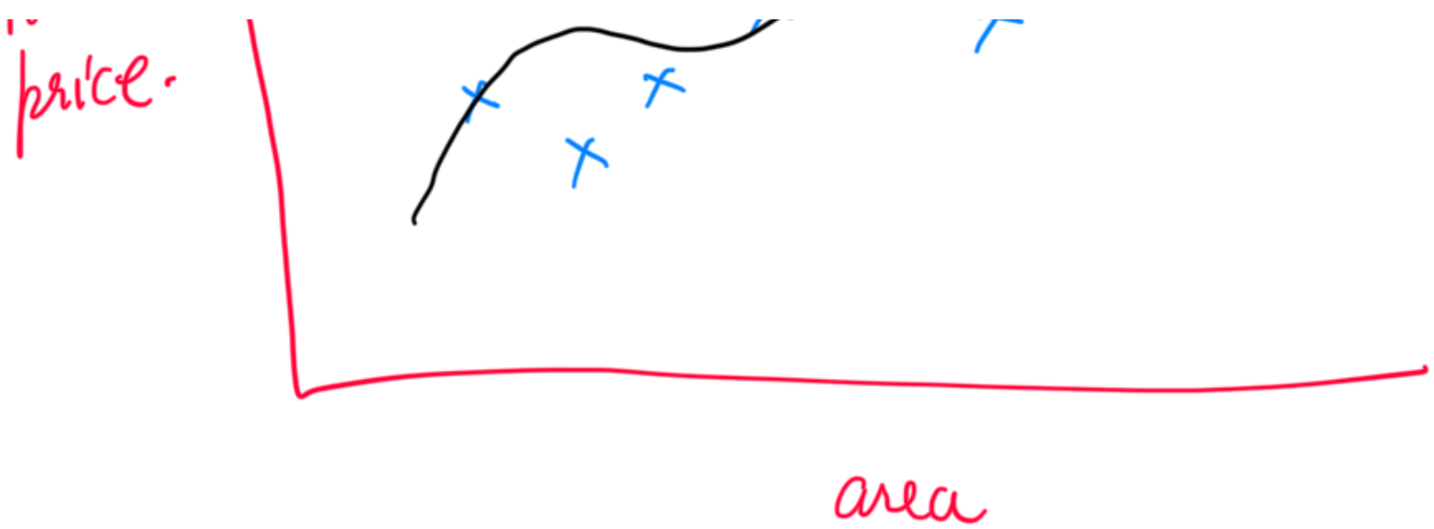
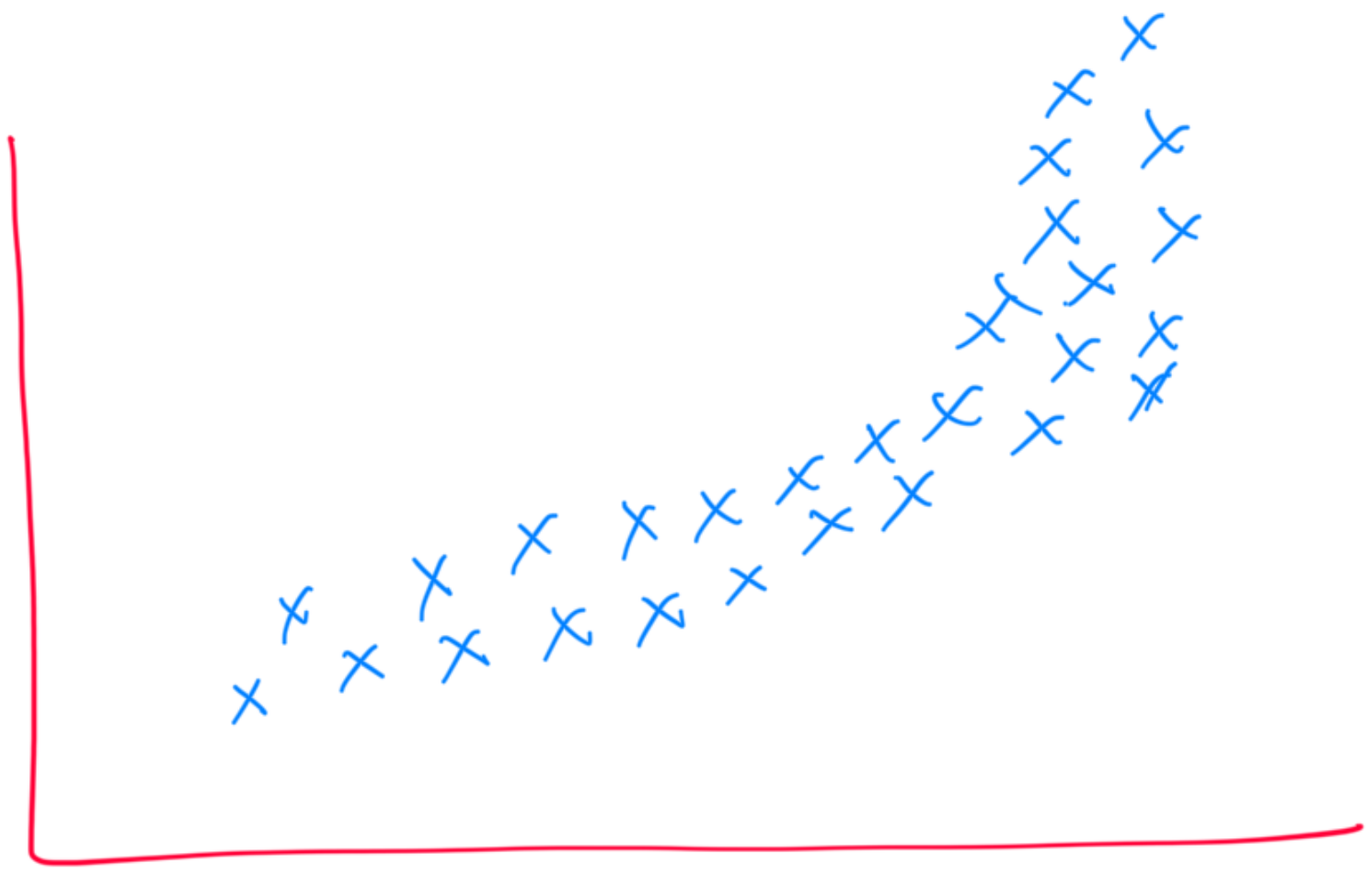
	Training	Testing
good learning	90 %	90 %
overfitting (ratta)	90 %	60 %
underfitting	60 %	60 %

overfitting: using complicated model for simpler situation.





Polynomial regression

degree = 2

$f(x) = x^2 \rightarrow$ degree

degree $= 3$

$$f(x) = 5x^3 + 4x^2 + 2x + 1$$

area	no of room

area $\quad$ area^2 $\quad$ area^3

overfitting

Regularization

$$w_1 x_1 + w_2 x_2 + w_3 x_3$$

$$\underbrace{50}$$

40

Regularization

$\downarrow$

Ridge

Lasso

	Train	testing
degree 3	85 → 81	75 → 78
degree 2	70	69

Bias - variance trade-off ??